

1. Introduction

In this assignment, a plant classification system was developed to recognize different vegetation species on campus using image data. The goal was to build an end-to-end image classification pipeline using a traditional machine learning approach, as required by the assignment.

A dataset of campus plants was collected manually, ensuring variation in lighting, viewpoints, and backgrounds. The system was designed to classify images into 10 different plant categories.

Unlike deep learning approaches, this project uses feature extraction and a Support Vector Machine (SVM) classifier, which allows for interpretable and efficient classification with a relatively small dataset.

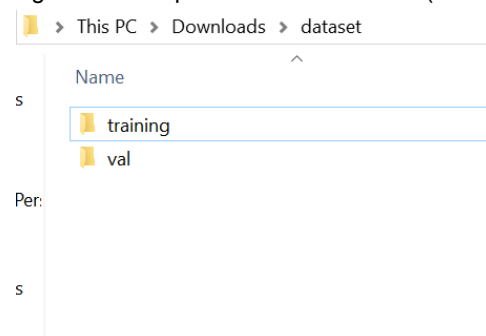
2. Dataset Collection and Organization

The dataset consists of ten plant classes: bougainvillea, cordyline_fruticosa, impatiens_walleriana, ixora, pentas_lanceolata, plumbago_auriculata, podocarpus_macrophyllus, raphiolepis_indica, schefflera_arboricola_variegated, and tibouchina.

Images were collected from multiple locations around campus. In many cases, plant name plates were used to confirm the ground-truth label, and additional images were taken from different angles and distances to capture variation within each class. This helped ensure that the model does not rely on a single viewpoint or background.

The dataset was split into a training set and a validation set. The training set contains 96 images and is used to train the model, while the validation set contains 29 images and is used to evaluate performance on unseen data.

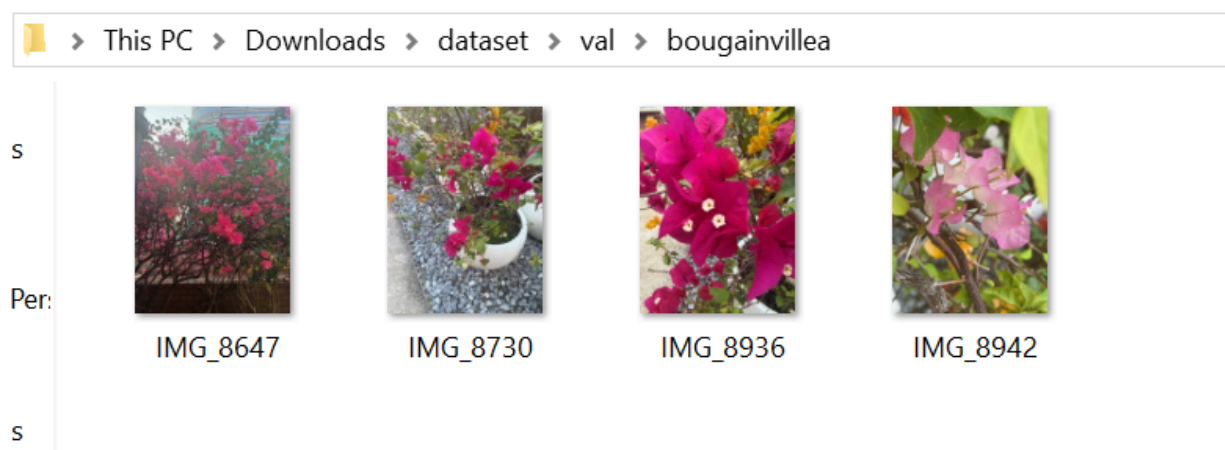
Figure 1: Example of dataset structure (training/validation folders with class subfolders)



This PC > Downloads > dataset > training			
Name	Date modified	Type	
tibouchina	3/22/2026 6:01 PM	File folder	
schefflera_arboricola_variegated	3/22/2026 5:57 PM	File folder	
raphiolepis_indica	3/22/2026 6:03 PM	File folder	
podocarpus_macrophyllus	3/22/2026 6:14 PM	File folder	
plumbago_auriculata	3/22/2026 6:13 PM	File folder	
pentas_lanceolata	3/22/2026 6:12 PM	File folder	
ixora	3/22/2026 6:10 PM	File folder	
impatiens_walleriana	3/22/2026 6:08 PM	File folder	
cordyline_fruticosa	3/22/2026 6:05 PM	File folder	
bougainvillea	3/22/2026 5:52 PM	File folder	

This PC > Downloads > dataset > val			
Name	Date modified	Type	
bougainvillea	3/22/2026 5:53 PM	File folder	
cordyline_fruticosa	3/22/2026 6:05 PM	File folder	
impatiens_walleriana	3/22/2026 6:08 PM	File folder	
ixora	3/22/2026 6:10 PM	File folder	
pentas_lanceolata	3/22/2026 6:12 PM	File folder	
plumbago_auriculata	3/22/2026 6:13 PM	File folder	
podocarpus_macrophyllus	3/22/2026 6:14 PM	File folder	
raphiolepis_indica	3/22/2026 6:03 PM	File folder	
schefflera_arboricola_variegated	3/22/2026 5:56 PM	File folder	
tibouchina	3/22/2026 6:01 PM	File folder	

Figure 2: Sample images from the bougainvillea class



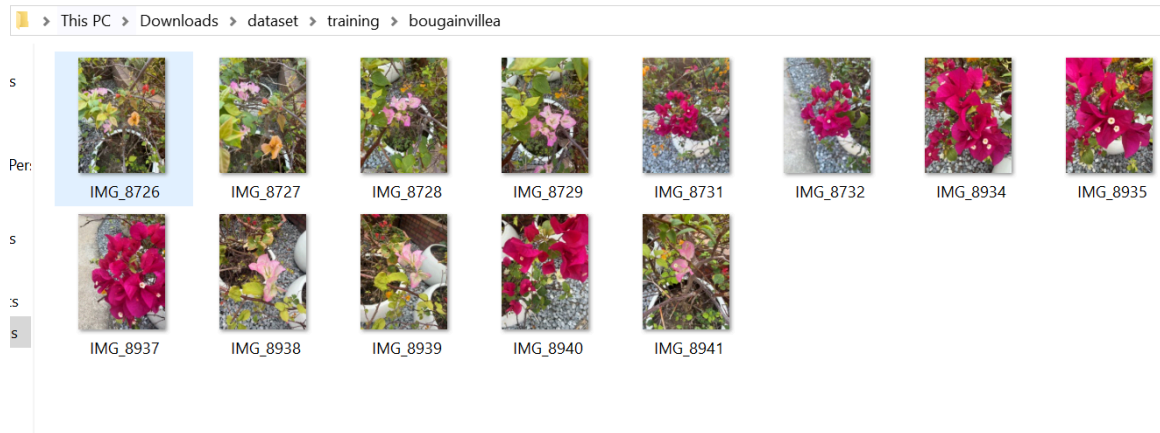


Figure 3: Plant name plates



3. Design, Testing, Robustness, and GenAI Discussion

The program was designed as a reproducible pipeline with four main steps: data loading and labeling with a fixed directory structure; preprocessing and feature extraction; model training; and, finally, evaluation and application inference. Each of these steps was individually tested before integrating all steps into the complete pipeline. For instance, I checked that loading images correctly produces the correct label list for all ten classes. I also checked that all images are correctly converted to feature vectors of the same length. This is important to ensure that the classifier gets the correct input. I then tested the end-to-end pipeline by training the SVM on the training set and evaluating performance on a held-out validation set using accuracy, confusion matrix, and per-class metrics.

Robustness was considered during both data collection and deployment. During the process of collecting the data, images were captured from different viewpoints, lighting conditions, and backgrounds to avoid relying too much on the scene context of the image. During the inference phase, the same preprocessing techniques that were applied during training are applied to the uploaded images to match the feature format of the real-world images with that of the model. Robustness was further checked using real-world test images taken on campus in more complex environments; these trials showed that performance decreases when the background is cluttered or multiple plants appear in the same image, which is expected for a color-histogram feature representation.

The only significant problem noticed was the confusion between plant classes with similar color distribution patterns (e.g., different types of pink flowering plants), and inaccuracy when there were shifts in the features of the histogram caused by changes in the lighting. This was overcome by converting images to HSV rather than keeping them in RGB format because of the stability in such images when exposed to different lighting conditions, and also by normalizing the histograms to ensure that the magnitudes of the features were similar. In addition, I included qualitative testing using real-world photos to identify typical failure cases such as background bias and multi-object scenes, which helped confirm the limitations of color-based features.

GenAI was used as a coding assistant during development. It helped generate initial code templates for feature extraction (HSV conversion and histogram computation), evaluation utilities (confusion matrix and classification report generation), and the upload-based prediction interface for the application. It was also used for debugging, such as identifying why certain plotting code did not apply to an SVM pipeline and suggesting the correct evaluation-based outputs for traditional ML. All GenAI-generated code was manually reviewed and tested in the notebook, and only retained when it produced correct outputs consistent with the assignment requirements.

GenAI assists by mapping a natural language request (e.g., “compute a confusion matrix for this classifier” or “build an upload-and-predict interface”) to common programming patterns it has learned from training on large corpora of technical text and code. It proposes candidate implementations based on the identified patterns, and the user refines the solution by providing constraints like “deep learning not allowed” or “use SVM,” and also validates the output by executing it.

The GenAI can arrive at a solution that appears correct but does not conform to the task constraints (for instance, proposing a deep learning training curve or neural network where traditional ML is specified). It can also leave out important details that are necessary for reproducibility (such as a random seed value, documentation of parameters used, etc.). To ensure accuracy in GenAI code proposals, it is recommended that they be reviewed as drafts. The user should check all code for accuracy against the assignment rubric, and explicit documentation should be added to ensure accuracy in the final product.

4. Methodology

Since traditional machine learning models cannot directly process raw images, a feature extraction step was required. In this project, each image was converted into a numerical representation using a color histogram.

First, all images were resized to a fixed size to ensure consistency. They were then converted from RGB to HSV, which is a more suitable color space for dealing with different

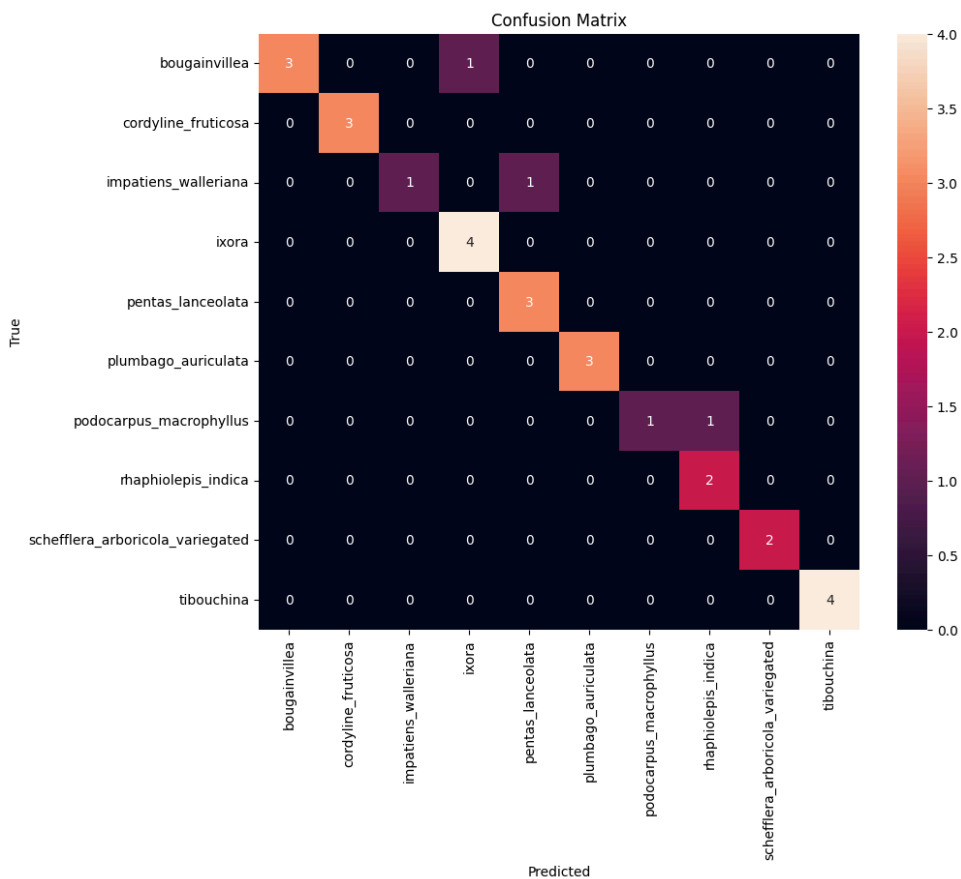
lighting conditions. A histogram was created to capture the colors present in each of the images. The histogram was normalized, creating a feature vector that represents the images.

These feature vectors were used to train a Support Vector Machine classifier with an RBF kernel. The reason behind choosing the SVM classifier was that it performs well on small datasets and is capable of handling non-linear classification boundaries. It is also an efficient classifier and suitable for the feature representation used in this project.

5. Model Evaluation

The trained model has been tested on the validation set. The accuracy of the model on the validation set is approximately 90%, indicating strong performance on the held-out dataset. Because the held-out set is small (29 images) with low per-class support, this accuracy should be interpreted cautiously; therefore, I also report per-class precision/recall/F1 and analyze typical failure modes to better characterize generalization.

Figure 4: Confusion Matrix



From the confusion matrix above, it is noted that some of the plant classes have been correctly classified. For example, ixora and tibouchina plants have been correctly classified most of the time. However, there are instances where some of the plants have been wrongly classified. For example, some of the flowering plants have been wrongly classified as other similar plants. Similarly, some green foliage plants have also been incorrectly classified based on similar color distribution.

In addition to the confusion matrix, a classification report was also created to obtain precision, recall, and F1-score values for all classes. This would provide a detailed overview of how well the model is performing on different plant classes. Precision measures how well the model has correctly predicted the instances of a given class. Similarly, recall measures how well the model has correctly identified the instances of a given class. The F1-score is a combination of precision and recall.

Figure 5: Classification Report

*** Validation Accuracy: 0.896551724137931

Classification Report:

	precision	recall	f1-score	support
bougainvillea	1.00	0.75	0.86	4
cordyline_fruticosa	1.00	1.00	1.00	3
impatiens_walleriana	1.00	0.50	0.67	2
ixora	0.80	1.00	0.89	4
pentas_lanceolata	0.75	1.00	0.86	3
plumbago_auriculata	1.00	1.00	1.00	3
podocarpus_macrophyllus	1.00	0.50	0.67	2
raphiolepis_indica	0.67	1.00	0.80	2
schefflera_arboricola_variegated	1.00	1.00	1.00	2
tibouchina	1.00	1.00	1.00	4
accuracy			0.90	29
macro avg	0.92	0.88	0.87	29
weighted avg	0.92	0.90	0.89	29

6. Error Analysis

The main source of error in this model comes from the limitations of the feature extraction method. Since the model relies only on color histograms, it focuses primarily on color distribution and ignores important structural features such as leaf shape, texture, and spatial arrangement.

As a result, plants with similar colors but different shapes are often confused. For example, flowers such as impatiens and pentas that have similar colors are likely to be

confused with each other despite their differences in shape. Other plants, such as podocarpus and raphiolepis, that have similar colors are likely to be confused with each other.

In addition, the lighting conditions will have an impact on the performance as well. The change in the amount of light or the presence of a shadow will result in a change of distribution of colors, which will have a direct impact on the histogram. The background, which includes soil, pots, or other plants, will have an impact as well.

For example, some *impatiens_walleriana* images were misclassified as *pentas_lanceolata* due to similar pink color distributions, highlighting the limitation of color-based features.

7. Application (Task 5)

A simple application was implemented using Python in Google Colab. This application allows a user to upload an image and receive a predicted plant class along with a confidence score.

The uploaded image is processed using the same feature extraction method as the training data. The trained SVM model then predicts the class based on the extracted features.

To use the application, the user runs all cells in the notebook and executes the upload cell. A file selection interface appears, allowing the user to choose an image. The model then outputs the predicted class and confidence score.

Figure 6: Application output



8. Real-World Testing

To evaluate how the model performs in real-world scenarios, additional images were taken on campus, including images with plant name plates and more complex backgrounds.

Figure 7: Real-World Test Images



The model has been able to correctly identify some of the plants, like bougainvillea, with moderate confidence. However, the results were not consistent when there were multiple plants, cluttered backgrounds, and non-uniform lighting. This shows that the model is domain-sensitive, where real-world images differ from the training set in terms of backgrounds and lighting conditions.

This demonstrates that while the model performs reasonably well on controlled data, it is less robust in real-world conditions where variability is higher.

9. Discussion

The model had moderate accuracy despite the limited dataset and basic feature set. This proves that traditional machine learning techniques can be used to solve image classification problems with suitable feature extraction.

However, the model has some disadvantages. The fact that it uses color features means it is not suitable to work with changes in lighting and is not able to recognize structural changes in plants. The model would perform better with a larger dataset and the addition of new features like texture and shape. The use of advanced machine learning techniques like deep learning would give better accuracy, but this is not allowed in this assignment.

10. Conclusion

In conclusion, a conventional machine learning pipeline has been successfully implemented with the use of color histogram features and the SVM classifier for campus

vegetation classification. The model has shown its capability to learn from the data and achieve accuracy for unseen data.

The application allows users to interact with the model by uploading images and receiving predictions, making the system practical and easy to use. While there are limitations in performance, this project provides a strong foundation for further improvements in plant classification.